Clinical Evidence for the Construction of the CliniCause Representation

Supplementary Material

Abstract

This standalone supplement documents the evidence trail for the deterministic proxy exposures and dataset-specific directed acyclic graphs (DAGs) used by CliniCause. It covers exactly 19 admitted proxy exposures (nine MIMIC-III and ten PhysioNet 2012) and all 102 implemented graph edges (57 and 45, respectively). The tables separate preserved ChatGPT source provenance, current deterministic implementation, later independent verification, and post-construction corroboration. Missing evidence is reported explicitly. The document does not claim that every local threshold or directed edge is literature validated.

1 Purpose and evidence boundary

The binary variables documented here are analytical proxy constructs, not chart-adjudicated diagnoses. They apply inspectable, deterministic rules to available measurements and care-process indicators. A positive tag therefore means that the implemented evidence rule fired; it does not establish that a clinician diagnosed the condition.

The DAGs are similarly inspectable structural hypotheses. Their arrows encode the project's assumed ordering among background variables, proxy states, observations, care processes, measurement processes, and in-hospital mortality. They are neither learned causal truth nor a completed clinician-adjudicated graph.

Four evidence layers are kept separate:

1. **Original ChatGPT evidence**: a source receives this label only when a preserved final response explicitly contains the source and the stated mapping.
2. **Current deterministic implementation**: establishes current names, rules, modes, nodes, and edges, but not clinical validity or original provenance.
3. **Later independent verification**: locally available papers inspected after construction. These sources are marked as new verification evidence and are never relabeled as original ChatGPT sources.
4. **Post-construction corroboration**: later comparisons of estimated mortality directions; these are excluded from construction evidence.

PhysioNet rules use the default threshold dictionary and whole first-48-hour record summaries. MIMIC rules currently default to canonical-pickle input and use whole-record extrema/first values and any-flags, with explicit first-24-hour and rolling-six-hour urine summaries. Current-source inspection does not by itself prove the exact historical input artifact or source revision that produced every archived resource.

2 Evidence-status legend

Provenance labels

Confirmed exact original ChatGPT mapping The final export places the source with the named proxy rationale. This says nothing about whether the source validates the exact local composite.

Confirmed family-level original ChatGPT source The final export maps the source to a clinical or observation-process family, not an individual final edge.

Source present; exact mapping not recoverable The source appears in the final export, but proximity is insufficient to claim an exact mapping.

No original ChatGPT source found No proxy- or edge-specific source was recovered from the preserved final response.

Verification labels

Verified—supports stated claim The locally inspected full text supports the bounded statement and a passage locator is recorded.

Verified—partially supports stated claim Only part of the measurement/rule statement is supported.

Verified—supports broad construct, not exact cutoff The source supports the construct or standard definition, but not the complete local score.

Verified—supports association, not directed causality The source supports an association or predictive signal, not the direction of a graph arrow.

Not verified Full text was unavailable, no specific source was mapped, or the source was not inspected.

Source identifiers beginning with `O_` are preserved original-export sources. Identifiers beginning with `V_` are **new independent-verification sources—not part of the original ChatGPT elicitation**. In particular, `O_SOFA_MSD`, `O_BERLIN_MSD`, and `O_KDIGO_MSD` are the derivative web tables actually linked by the final MIMIC response; the corresponding original SOFA, Berlin, and KDIGO full texts were inspected later and carry separate `V_` identifiers. `O_SEPSIS3` is both an original-export paper and a locally inspected full text.

3 Proxy-definition evidence table

The following is one logical proxy-definition table, continued by dataset. Chronic baseline constructs are described in the graph but are not counted as admitted exposures. Each primary row is followed by its inputs, rule/window, independent support paraphrase, and qualification. Full clause-level audit data are retained in the accompanying CSV.

Table 1: Proxy-definition evidence: 19 admitted proxy exposures.

Proxy, current rule, provenance, independent verification, and qualification
M01 — `LAT_INFLAMMATION_SEPSIS` *Dataset:* MIMIC-III **Clinical interpretation:** Inflammation or suspected-infection/sepsis burden; not confirmed sepsis. **Original ChatGPT source IDs:** `O_SEPSIS3`. **DOI/stable URL:** `doi:10.1001/jama.2016.0287`. **Verdict:** VERIFIED — SUPPORTS THE BROAD CONSTRUCT, NOT THE EXACT CUTOFF. **Inputs:** Temperature_min/max; WBC_min/max; Lactate_max; CultureOrdered_any; CulturePositive_any; SuspectedInfection_any; AntibioticStarted_any/Antibiotics_any. **Rule/window:** Five indicators: temperature >=38.3 or <36; WBC >12 or <4; lactate >2; culture/suspected-infection; antibiotic/suspected-infection. Positive if score >=2 and a culture, antibiotic, temperature or WBC anchor is present. *Aggregation:* Whole available ICU record summaries; urine sum at 0-24h and minimum rolling 6h mL/kg/h; missing clauses do not fire. **Independent support:** Sepsis-3 supports infection-related dysregulated host response and organ dysfunction, but not this five-clause score. **Qualification:** Temperature/WBC are nonspecific; treatment and culture fields encode clinician suspicion; the exact score is not validated.
M02 — `LAT_GLOBAL_SEVERITY` *Dataset:* MIMIC-III **Clinical interpretation:** Multi-domain acute physiological severity. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** `https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score`; `doi:10.1007/BF01709751`. **Verdict:** VERIFIED — SUPPORTS THE BROAD CONSTRUCT, NOT THE EXACT CUTOFF. **Inputs:** MAP_min; SBP_min; Vasopressors_any; SpO2_min; PF_ratio_min; MechanicalVentilation_any; GCS_min; RASS_min/max; Creatinine_max; UrineOutput_sum_24h; Lactate_max; pH_min; Bicarbonate_min; Temperature_min/max; WBC_min/max; Bilirubin_max; Platelets_min; INR_max. **Rule/window:** Seven Boolean domains (circulatory, respiratory, neurologic, renal, metabolic, inflammatory, hepatic/coagulation); positive if at least 3 domains are abnormal. *Aggregation:* Whole available ICU record summaries; urine sum at 0-24h and minimum rolling 6h mL/kg/h; missing clauses do not fire. **Independent support:** The original source presents SOFA organ domains. The original SOFA paper verifies six domains and daily worst values, not the local seven-domain Boolean composite. **Qualification:** Local substitutions, inflammatory domain and score >=3 are project choices; whole-record aggregation differs from daily SOFA scoring.
M03 — `LAT_SHOCK` *Dataset:* MIMIC-III **Clinical interpretation:** Circulatory shock or clinically meaningful hemodynamic instability. **Original ChatGPT source IDs:** `O_SEPSIS3`; `O_SOFA_MSD`. **DOI/stable URL:** `doi:10.1001/jama.2016.0287`; `https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score`. **Verdict:** VERIFIED — SUPPORTS THE BROAD CONSTRUCT, NOT THE EXACT CUTOFF. **Inputs:** MAP_min; SBP_min; MAP_sustained_lt70_any; Vasopressors_any; Lactate_max; UrineOutput_sum_24h; UrineOutput_mlkg_6h_min; pH_min; BaseExcess_min. **Rule/window:** Five indicators: hypotension (MAP <65, SBP <=90 or sustained MAP <70), vasopressor, lactate >2, oliguria (<0.5 mL/kg/h over 6h or <500 mL/24h), acidosis (pH <7.30 or base excess <=-5). Positive if score >=2, or vasopressor plus hypoperfusion/acidosis/hypotension. *Aggregation:* Whole available ICU record summaries; urine sum at 0-24h and minimum rolling 6h mL/kg/h; missing clauses do not fire. **Independent support:** Sepsis-3 supports vasopressor-dependent hypotension with lactate >2 for septic shock; it does not support the project's any-two composite. **Qualification:** Rule is broader than septic shock and mixes treatment, perfusion, urine and acid-base evidence.

Table 1 continued

Proxy, current rule, provenance, independent verification, and qualification
M04 — `LAT_RESPIRATORY_FAILURE` *Dataset:* MIMIC-III **Clinical interpretation:** Hypoxemic or ventilatory failure and/or substantial respiratory support. **Original ChatGPT source IDs:** `O_BERLIN_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/berlin-definition-of-ards; doi:10.1001/jama.2012.5669. **Verdict:** VERIFIED — SUPPORTS THE BROAD CONSTRUCT, NOT THE EXACT CUTOFF. **Inputs:** SpO2_min; PF_ratio_min; FiO2_max; MechanicalVentilation_any; NonInvasiveVentilation_any; RR_min/max; PaCO2_max; pH_min. **Rule/window:** Five indicators: SpO2 <92; P/F <=300; respiratory support or FiO2 >=0.5; RR >=30 or <=8; PaCO2 >=50 with pH <7.30. Positive if score >=2, or support plus hypoxemia/PF/ventilatory failure. *Aggregation:* Whole available ICU record summaries; urine sum at 0-24h and minimum rolling 6h mL/kg/h; missing clauses do not fire. **Independent support:** Berlin supports P/F severity categories under PEEP/CPAP plus timing, imaging and edema-origin requirements; the project proxy is broader. **Qualification:** Not an ARDS diagnosis; several Berlin criteria are absent and ventilation may reflect treatment/case mix.
M05 — `LAT_RENAL_DYSFUNCTION` *Dataset:* MIMIC-III **Clinical interpretation:** Acute or clinically important renal dysfunction. **Original ChatGPT source IDs:** `O_KDIGO_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/staging-criteria-for-acute-kidney-injury-kdigo-2012; https://kdigo.org/guidelines/acute-kidney-injury/. **Verdict:** VERIFIED — SUPPORTS THE BROAD CONSTRUCT, NOT THE EXACT CUTOFF. **Inputs:** Creatinine_max/delta; BUN_max; UrineOutput_sum_24h; UrineOutput_mlkg_6h_min; Potassium_max; Bicarbonate_min; DialysisOrCRRT_any. **Rule/window:** Five indicators: creatinine >=2 or rise >=0.3; BUN >=40; oliguria (<0.5 mL/kg/h over 6h or <500 mL/24h); K >=5.5 or HCO3 <18; dialysis. Positive if score >=2 or dialysis. *Aggregation:* Whole available ICU record summaries; urine sum at 0-24h and minimum rolling 6h mL/kg/h; missing clauses do not fire. **Independent support:** KDIGO supports a >=0.3 mg/dL creatinine rise, baseline-relative changes and urine-output criteria; it does not validate this composite or BUN/electrolyte clauses. **Qualification:** Baseline creatinine may be unavailable; absolute creatinine, BUN and metabolic additions are project choices.
M06 — `LAT_HEPATIC_COAG_DYSFUNCTION` *Dataset:* MIMIC-III **Clinical interpretation:** Combined hepatic injury/reserve and coagulation dysfunction. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score; doi:10.1007/BF01709751. **Verdict:** VERIFIED — SUPPORTS THE BROAD CONSTRUCT, NOT THE EXACT CUTOFF. **Inputs:** Bilirubin_max; AST_max; ALT_max; Platelets_min; INR_max; PTProlonged_any; PTTProlonged_any; Albumin_min/first. **Rule/window:** Five indicators: bilirubin >=2; AST or ALT >=120; platelets <100; INR >=1.5 or prolonged PT/PTT; albumin <2.5. Positive if score >=2. *Aggregation:* Whole available ICU record summaries; urine sum at 0-24h and minimum rolling 6h mL/kg/h; missing clauses do not fire. **Independent support:** SOFA supports bilirubin and platelets as separate organ domains; ISTH DIC uses a different multi-test algorithm. Neither validates the combined local score. **Qualification:** Combines distinct constructs; anticoagulation, chronic liver disease, dilution and nutrition can trigger clauses.

Table 1 continued

Proxy, current rule, provenance, independent verification, and qualification
M07 — `LAT_NEUROLOGIC_DYSFUNCTION` *Dataset:* MIMIC-III **Clinical interpretation:** Depressed consciousness or focal/behavioral neurological dysfunction. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** doi:10.1007/BF01709751. **Verdict:** VERIFIED — SUPPORTS THE BROAD CONSTRUCT, NOT THE EXACT CUTOFF. **Inputs:** GCS_min; GCSComponentAbnormal_any; RASS_min/max; PupilOrFocalNeuroAbnormal_any; SedationOrIntubation_any. **Rule/window:** GCS <=8 contributes 2; else GCS <=13 contributes 1; abnormal GCS component, RASS <=-3 or >=2, and pupil/focal abnormality each contribute 1. Positive if score >=2; sedation/intubation with an isolated one-point score is forced negative. *Aggregation:* Whole available ICU record summaries; urine sum at 0-24h and minimum rolling 6h mL/kg/h; missing clauses do not fire. **Independent support:** SOFA uses GCS as its neurological domain, but does not support the RASS, focal-sign or sedation-exclusion composite. **Qualification:** Sedation/intubation can confound GCS/RASS; exact weighting and exclusions lack direct literature validation.
M08 — `LAT_METABOLIC_DERANGEMENT` *Dataset:* MIMIC-III **Clinical interpretation:** Acid-base, lactate, electrolyte or glucose derangement. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Inputs:** pH_min/max; Bicarbonate_min/max; BaseExcess_min; Lactate_max; Potassium_min/max; Sodium_min/max; Glucose_min/max. **Rule/window:** Six domains: pH <7.30 or >7.55; HCO3 <18 or >35 or base excess <=-5; lactate >2; K <3 or >=5.5; Na <130 or >150; glucose <70 or >250. Positive if at least 2 domains are abnormal. *Aggregation:* Whole available ICU record summaries; urine sum at 0-24h and minimum rolling 6h mL/kg/h; missing clauses do not fire. **Independent support:** No locally inspected paper validates the exact six-domain score. **Qualification:** Broad construct is plausible but all cutoffs and the any-two combination remain unverified.
M09 — `LAT_CARDIAC_STRAIN` *Dataset:* MIMIC-III **Clinical interpretation:** Myocardial injury, rhythm instability or severe cardiac strain. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** doi:10.1161/CIR.0000000000000617. **Verdict:** NOT VERIFIED — FULL TEXT UNAVAILABLE. **Inputs:** TroponinPositive/AboveULN flags; TroponinT_max; TroponinI_max; CKMB_max/AboveULN; Arrhythmia_any; HR_min/max; MAP_min; SBP_min; FirstCareUnit; CardiacSurgeryContext_any. **Rule/window:** Five indicators: troponin abnormal (or T >=0.1/I >=0.4 fallback); CK-MB abnormal; arrhythmia or HR >150/<40; HR >130 or <40 with hypotension; CCU/CSRU or cardiac-surgery context. Positive if score >=2 and troponin or rhythm anchor. *Aggregation:* Whole available ICU record summaries; urine sum at 0-24h and minimum rolling 6h mL/kg/h; missing clauses do not fire. **Independent support:** Phase 2 identified myocardial-injury sources, but their full text was not locally available and they were not original ChatGPT sources. **Qualification:** Assay-specific troponin thresholds, CK-MB >0, unit/care-context points and the composite are not verified.

Table 1 continued

Proxy, current rule, provenance, independent verification, and qualification

P01 — `LAT_GLOBAL_SEVERITY` *Dataset:* PhysioNet 2012

Clinical interpretation: Multi-domain first-48h acute physiological severity. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score; doi:10.1007/BF01709751. **Verdict:** VERIFIED — SUPPORTS THE BROAD CONSTRUCT, NOT THE EXACT CUTOFF.
Inputs: MAP/NIMAP/SysABP/NISysABP__min; MechVent__max; PF__min or PaO2/FiO2 approximation; SaO2__min; RespRate__max; GCS__min; Creatinine/BUN__max; urine summary; Bilirubin__max; Platelets__min; Lactate__max; pH__min; HCO3__min; TropI/T__max; HR__max.
Rule/window: Seven domains (hemodynamic, respiratory, neurologic, renal, hepatic/coagulation, metabolic, cardiac). Positive if score >=3, or score >=2 plus lactate >=4, pH <7.20, ventilation, GCS <=8 or MAP <60. *Aggregation:* Whole first-48h Challenge record; min/max/mean/first/last summaries; missing clauses do not fire.
Independent support: SOFA supports six organ domains and published ordinal thresholds, not the project's seven Boolean domains or critical override. **Qualification:** Adds cardiac/metabolic substitutions and uses whole-48h summaries rather than daily SOFA scoring.

P02 — `LAT_SHOCK` *Dataset:* PhysioNet 2012

Clinical interpretation: Circulatory shock or hemodynamic instability without vasopressor data. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score; doi:10.1007/BF01709751. **Verdict:** VERIFIED — SUPPORTS THE BROAD CONSTRUCT, NOT THE EXACT CUTOFF.
Inputs: MAP/NIMAP/SysABP/NISysABP__min; Lactate__max; HR__max; urine summary; pH__min; HCO3__min.
Rule/window: Six indicators: MAP/NIMAP <70; SBP/NISBP <=90; lactate >2; HR >=110; urine <500; pH <7.30 or HCO3 <18. Positive if score >=2 or low MAP with lactate >=4. *Aggregation:* Whole first-48h Challenge record; min/max/mean/first/last summaries; missing clauses do not fire.
Independent support: SOFA supports MAP <70 as cardiovascular dysfunction; it does not support the complete any-two score or lactate override. **Qualification:** No vasopressor field; low BP, tachycardia, oliguria and acidosis are nonspecific.

P03 — `LAT_RESPIRATORY_FAILURE` *Dataset:* PhysioNet 2012

Clinical interpretation: Hypoxemic or ventilatory respiratory failure/support. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score; doi:10.1001/jama.2012.5669. **Verdict:** VERIFIED — SUPPORTS THE BROAD CONSTRUCT, NOT THE EXACT CUTOFF.
Inputs: MechVent__max; PF__min or PaO2/FiO2 approximation; SaO2__min; PaO2__min; RespRate__min/max; PaCO2__max; pH__min.
Rule/window: Five indicators: ventilation; P/F <300; SaO2 <92 or PaO2 <60; RR >=22 or <8; PaCO2 >=50 or PaCO2 >=45 with pH <7.30. Positive if score >=2 or ventilation plus P/F <300. *Aggregation:* Whole first-48h Challenge record; min/max/mean/first/last summaries; missing clauses do not fire.
Independent support: SOFA supports P/F oxygenation scoring; Berlin and Essay support narrower respiratory/ventilation constructs, not this full composite. **Qualification:** P/F may be approximated from non-paired extrema; not an ARDS diagnosis.

Table 1 continued

Proxy, current rule, provenance, independent verification, and qualification
P04 — `LAT_RENAL_DYSFUNCTION` *Dataset:* PhysioNet 2012 **Clinical interpretation:** Acute or clinically important renal dysfunction. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score; https://kdigo.org/guidelines/acute-kidney-injury/. **Verdict:** VERIFIED — SUPPORTS THE BROAD CONSTRUCT, NOT THE EXACT CUTOFF. **Inputs:** Creatinine_first/max; BUN_max; urine summary; K_max; HCO3_min. **Rule/window:** Five indicators: creatinine >=2; creatinine rise >=0.3; BUN >=40; urine <500; K >=5.5 or HCO3 <18. Positive if score >=2 or creatinine >=3.5. *Aggregation:* Whole first-48h Challenge record; min/max/mean/first/last summaries; missing clauses do not fire. **Independent support:** SOFA supports creatinine/urine domains; KDIGO supports >=0.3 rise and timed urine criteria, not the whole local score. **Qualification:** Current urine helper accepts precomputed summary names that the simple summarizer does not itself create; baseline CKD is unresolved.
P05 — `LAT_HEPATIC_DYSFUNCTION` *Dataset:* PhysioNet 2012 **Clinical interpretation:** Cholestatic/hepatocellular injury or impaired hepatic reserve. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score; doi:10.1007/BF01709751. **Verdict:** VERIFIED — SUPPORTS THE BROAD CONSTRUCT, NOT THE EXACT CUTOFF. **Inputs:** Bilirubin_max; AST_max; ALT_max; ALP_max; Albumin_min; Platelets_min. **Rule/window:** Weighted score: bilirubin >=2 contributes 2; AST/ALT >=200, ALP >=250, albumin <2.5 and platelets <100 each contribute 1. Positive if score >=2. *Aggregation:* Whole first-48h Challenge record; min/max/mean/first/last summaries; missing clauses do not fire. **Independent support:** SOFA supports bilirubin as a liver-domain marker; the added enzymes, albumin and platelet weighting are not the SOFA definition. **Qualification:** Chronic liver disease, hemolysis and acute inflammation can trigger clauses; exact composite unsupported.
P06 — `LAT_COAG_HEME_DYSFUNCTION` *Dataset:* PhysioNet 2012 **Clinical interpretation:** Thrombocytopenia, anemia/polycythemia or hematologic stress. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score; doi:10.1055/s-0037-1616068. **Verdict:** VERIFIED — SUPPORTS THE BROAD CONSTRUCT, NOT THE EXACT CUTOFF. **Inputs:** Platelets_first/min; HCT_min/max; WBC_min/max. **Rule/window:** Weighted score: platelets <100 contributes 2; platelets <150, HCT <25/>55, WBC <4/>20, and platelet drop >=50 each contribute 1. Positive if score >=2. *Aggregation:* Whole first-48h Challenge record; min/max/mean/first/last summaries; missing clauses do not fire. **Independent support:** SOFA supports platelet thresholds; ISTH DIC uses platelets with PT, fibrin-related markers and fibrinogen, not HCT/WBC additions. **Qualification:** Not a DIC diagnosis; overlapping platelet weights and non-coagulation HCT/WBC clauses are project choices.

Table 1 continued

Proxy, current rule, provenance, independent verification, and qualification
P07 — `LAT_INFLAMMATION_SEPSIS_BURDEN` *Dataset:* PhysioNet 2012 **Clinical interpretation:** Systemic inflammatory or sepsis-like host response; not confirmed sepsis. **Original ChatGPT source IDs:** `O_SEPSIS3`. **DOI/stable URL:** `doi:10.1001/jama.2016.0287`. **Verdict:** VERIFIED — SUPPORTS THE BROAD CONSTRUCT, NOT THE EXACT CUTOFF. **Inputs:** Temp_min/max; WBC_min/max; HR_max; RespRate_max; PaCO2_min; Lactate_max; Platelets_min. **Rule/window:** Six indicators: temperature >38.3 or <36; WBC >12 or <4; HR >90; RR >20 or PaCO2 <32; lactate >2; platelets <150. Positive if score >=3, or temperature/WBC abnormal plus lactate and score >=2. *Aggregation:* Whole first-48h Challenge record; min/max/mean/first/last summaries; missing clauses do not fire. **Independent support:** Sepsis-3 requires infection-related organ dysfunction and explicitly supersedes a simple inflammation-only framing; the local dataset lacks an infection anchor. **Qualification:** False positives include sterile inflammation; exact SIRS-like score is not Sepsis-3.
P08 — `LAT_NEUROLOGIC_DYSFUNCTION` *Dataset:* PhysioNet 2012 **Clinical interpretation:** Depressed consciousness or neurological dysfunction with metabolic/gas contributors. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** `https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score`; `doi:10.1007/BF01709751`. **Verdict:** VERIFIED — SUPPORTS THE BROAD CONSTRUCT, NOT THE EXACT CUTOFF. **Inputs:** GCS_min; Na_min/max; Glucose_min/max; PaCO2_max; SaO2_min; pH_min. **Rule/window:** GCS <=12 contributes 2; GCS <15, sodium <130/>150 or glucose <70/>300, and PaCO2 >=50/SaO2 <90/pH <7.25 each contribute 1. Positive if score >=2. *Aggregation:* Whole first-48h Challenge record; min/max/mean/first/last summaries; missing clauses do not fire. **Independent support:** SOFA supports GCS as a neurological domain; it does not support adding electrolyte/gas clauses to the same weighted proxy. **Qualification:** Sedation/ventilation context is not used by the current function; metabolic/gas abnormalities can create nonspecific positives.
P09 — `LAT_CARDIAC_INJURY_STRAIN` *Dataset:* PhysioNet 2012 **Clinical interpretation:** Myocardial injury, ischemia or severe cardiac strain. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** `doi:10.1161/CIR.0000000000000617`. **Verdict:** NOT VERIFIED — FULL TEXT UNAVAILABLE. **Inputs:** TropI/T_max; ICUType_first; HR_min/max; MAP_min; SysABP_min; Lactate_max. **Rule/window:** Weighted score: troponin I >0.1 or T >0.01 contributes 2; ICU type 1/2, HR >=130/<50, MAP <70 or SBP <=90, and lactate >2 each contribute 1. Positive if score >=2. *Aggregation:* Whole first-48h Challenge record; min/max/mean/first/last summaries; missing clauses do not fire. **Independent support:** Phase 2 identified myocardial-injury sources, but no local full text was available and no original source was mapped. **Qualification:** Troponin assay thresholds are not harmonized; ICU type can make the proxy positive with one additional nonspecific signal.

Table 1 continued

Proxy, current rule, provenance, independent verification, and qualification
P10 — `LAT_METABOLIC_DERANGEMENT` *Dataset:* PhysioNet 2012 **Clinical interpretation:** Acid-base, lactate, electrolyte or glucose derangement. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Inputs:** pH_min/max; HCO3_min/max; Lactate_max; Na_min/max; K_min/max; Mg_min/max; Glucose_min/max; PaCO2_min/max. **Rule/window:** Six domains: pH <7.30/>7.50; HCO3 <18/>32; lactate >2; Na <130/>150, K <3/>5.5 or Mg <0.6/>1.2; glucose <70/>250; PaCO2 <32/>50. Positive if score >=2 or pH <7.20, lactate >=4 or K >=6. *Aggregation:* Whole first-48h Challenge record; min/max/mean/first/last summaries; missing clauses do not fire. **Independent support:** No locally inspected paper validates this exact multi-domain score or critical overrides. **Qualification:** Cutoffs, domain grouping and sufficient conditions are project choices.

4 DAG-edge evidence table

The machine-readable edge inventories are the structural authority. The tables below preserve exact node names and directions. A family-level source is not an edge-specific source. Most arrows therefore remain unverified as directed causal relationships even when a paper supports the associated construct, measurement, or mortality association.

4.1 MIMIC-III: 57 directed edges

Table 2: MIMIC-III DAG-edge evidence (57 edges).

Edge, rationale, provenance, independent check, and qualification
M01: `BG_ADMISSION_CONTEXT` → `LAT_GLOBAL_SEVERITY` *Dataset:* MIMIC-III **Rationale:** Project assumption: baseline/case-mix context precedes and influences the downstream proxy state. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M02: `BG_ADMISSION_CONTEXT` → `LAT_INFLAMMATION_SEPSIS` *Dataset:* MIMIC-III **Rationale:** Project assumption: baseline/case-mix context precedes and influences the downstream proxy state. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M03: `BG_ADMISSION_CONTEXT` → `MISS_MEASUREMENT_INTENSITY` *Dataset:* MIMIC-III **Rationale:** Project assumption: case mix and care setting influence monitoring or test-ordering intensity. **Original ChatGPT source IDs:** `O_SISK2021`. **DOI/stable URL:** `doi:10.1093/jamia/ocaa242`. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M04: `BG_AGE` → `LAT_CHRONIC_BURDEN` *Dataset:* MIMIC-III **Rationale:** Project assumption: baseline/case-mix context precedes and influences the downstream proxy state. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M05: `BG_AGE` → `OUT_MORTALITY` *Dataset:* MIMIC-III **Rationale:** Project assumption: baseline risk can affect mortality through measured and unmeasured pathways. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.

Table 2 continued

Edge, rationale, provenance, independent check, and qualification
M06: `BG_ETHNICITY_INSURANCE_LANGUAGE` → `MISS_MEASUREMENT_INTENSITY` *Dataset:* MIMIC-III **Rationale:** Project assumption: case mix and care setting influence monitoring or test-ordering intensity. **Original ChatGPT source IDs:** `O_SISK2021`. **DOI/stable URL:** doi:10.1093/jamia/ocaa242. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M07: `BG_ETHNICITY_INSURANCE_LANGUAGE` → `OUT_MORTALITY` *Dataset:* MIMIC-III **Rationale:** Project assumption: baseline risk can affect mortality through measured and unmeasured pathways. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M08: `BG_ICU_UNIT` → `LAT_GLOBAL_SEVERITY` *Dataset:* MIMIC-III **Rationale:** Project assumption: baseline/case-mix context precedes and influences the downstream proxy state. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M09: `BG_ICU_UNIT` → `MISS_MEASUREMENT_INTENSITY` *Dataset:* MIMIC-III **Rationale:** Project assumption: case mix and care setting influence monitoring or test-ordering intensity. **Original ChatGPT source IDs:** `O_SISK2021`. **DOI/stable URL:** doi:10.1093/jamia/ocaa242. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M10: `BG_SEX` → `LAT_CHRONIC_BURDEN` *Dataset:* MIMIC-III **Rationale:** Project assumption: baseline/case-mix context precedes and influences the downstream proxy state. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.

Table 2 continued

Edge, rationale, provenance, independent check, and qualification
M11: `LAT_CARDIAC_STRAIN` → `LAT_SHOCK` *Dataset:* MIMIC-III **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M12: `LAT_CARDIAC_STRAIN` → `OBS_TROPONIN_ECG` *Dataset:* MIMIC-III **Rationale:** Measurement-model edge: the observed group is treated as a manifestation or measurement of the parent proxy state. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M13: `LAT_CARDIAC_STRAIN` → `OUT_MORTALITY` *Dataset:* MIMIC-III **Rationale:** Outcome edge: the parent state is assumed to represent pathophysiologic burden related to mortality. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M14: `LAT_CHRONIC_BURDEN` → `LAT_CARDIAC_STRAIN` *Dataset:* MIMIC-III **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M15: `LAT_CHRONIC_BURDEN` → `LAT_GLOBAL_SEVERITY` *Dataset:* MIMIC-III **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.

Table 2 continued

Edge, rationale, provenance, independent check, and qualification
M16: LAT_CHRONIC_BURDEN → LAT_RENAL_DYSFUNCTION *Dataset:* MIMIC-III **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M17: LAT_CHRONIC_BURDEN → OUT_MORTALITY *Dataset:* MIMIC-III **Rationale:** Outcome edge: the parent state is assumed to represent pathophysiologic burden related to mortality. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M18: LAT_GLOBAL_SEVERITY → LAT_METABOLIC_DERANGEMENT *Dataset:* MIMIC-III **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** O_SOFA_MSD. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M19: LAT_GLOBAL_SEVERITY → LAT_NEUROLOGIC_DYSFUNCTION *Dataset:* MIMIC-III **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** O_SOFA_MSD. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M20: LAT_GLOBAL_SEVERITY → LAT_RESPIRATORY_FAILURE *Dataset:* MIMIC-III **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** O_SOFA_MSD. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.

Table 2 continued

Edge, rationale, provenance, independent check, and qualification
M21: `LAT_GLOBAL_SEVERITY` → `MISS_MEASUREMENT_INTENSITY` *Dataset:* MIMIC-III **Rationale:** Observation-process edge: the parent state is assumed to influence selective ordering or monitoring. **Original ChatGPT source IDs:** `O_SISK2021`. **DOI/stable URL:** doi:10.1093/jamia/ocaa242. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M22: `LAT_GLOBAL_SEVERITY` → `OUT_MORTALITY` *Dataset:* MIMIC-III **Rationale:** Outcome edge: the parent state is assumed to represent pathophysiologic burden related to mortality. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** SOFA describes organ dysfunction and mortality association, not directed causality for this edge.
M23: `LAT_HEPATIC_COAG_DYSFUNCTION` → `OBS_BILIRUBIN_PLATELETS_INR` *Dataset:* MIMIC-III **Rationale:** Measurement-model edge: the observed group is treated as a manifestation or measurement of the parent proxy state. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** VERIFIED — PARTIALLY SUPPORTS THE STATED CLAIM. **Qualification:** Sources support selected laboratory domains, not the exact project grouping.
M24: `LAT_HEPATIC_COAG_DYSFUNCTION` → `OUT_MORTALITY` *Dataset:* MIMIC-III **Rationale:** Outcome edge: the parent state is assumed to represent pathophysiologic burden related to mortality. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** SOFA describes organ dysfunction and mortality association, not directed causality for this edge.
M25: `LAT_INFLAMMATION_SEPSIS` → `LAT_GLOBAL_SEVERITY` *Dataset:* MIMIC-III **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** `O_SEPSIS3`. **DOI/stable URL:** doi:10.1001/jama.2016.0287. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** Sepsis-3 supports infection-related organ dysfunction/shock concepts, not the exact directed project edge.

Table 2 continued

Edge, rationale, provenance, independent check, and qualification
M26: `LAT_INFLAMMATION_SEPSIS` → `LAT_HEPATIC_COAG_DYSFUNCTION` *Dataset:* MIMIC-III **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** `O_SEPSIS3`. **DOI/stable URL:** `doi:10.1001/jama.2016.0287`. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** Sepsis-3 supports infection-related organ dysfunction/shock concepts, not the exact directed project edge.
M27: `LAT_INFLAMMATION_SEPSIS` → `LAT_SHOCK` *Dataset:* MIMIC-III **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** `O_SEPSIS3`. **DOI/stable URL:** `doi:10.1001/jama.2016.0287`. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** Sepsis-3 supports infection-related organ dysfunction/shock concepts, not the exact directed project edge.
M28: `LAT_INFLAMMATION_SEPSIS` → `OBS_TEMP_WBC_CULTURES` *Dataset:* MIMIC-III **Rationale:** Measurement-model edge: the observed group is treated as a manifestation or measurement of the parent proxy state. **Original ChatGPT source IDs:** `O_SEPSIS3`. **DOI/stable URL:** `doi:10.1001/jama.2016.0287`. **Independent check/verdict:** VERIFIED — PARTIALLY SUPPORTS THE STATED CLAIM. **Qualification:** Sepsis-3 supports infection-related organ dysfunction/shock concepts, not the exact directed project edge.
M29: `LAT_INFLAMMATION_SEPSIS` → `OUT_MORTALITY` *Dataset:* MIMIC-III **Rationale:** Outcome edge: the parent state is assumed to represent pathophysiologic burden related to mortality. **Original ChatGPT source IDs:** `O_SEPSIS3`. **DOI/stable URL:** `doi:10.1001/jama.2016.0287`. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** Sepsis-3 supports infection-related organ dysfunction/shock concepts, not the exact directed project edge.
M30: `LAT_INFLAMMATION_SEPSIS` → `TRT_ANTIBIOTICS` *Dataset:* MIMIC-III **Rationale:** Care-process edge: evidence of the parent state is assumed to prompt or indicate the recorded treatment. **Original ChatGPT source IDs:** `O_SEPSIS3`. **DOI/stable URL:** `doi:10.1001/jama.2016.0287`. **Independent check/verdict:** VERIFIED — PARTIALLY SUPPORTS THE STATED CLAIM. **Qualification:** Sepsis-3 supports infection-related organ dysfunction/shock concepts, not the exact directed project edge.

Table 2 continued

Edge, rationale, provenance, independent check, and qualification
M31: `LAT_METABOLIC_DERANGEMENT` → `OBS_PH_ELECTROLYTES_GLUCOSE` *Dataset:* MIMIC-III **Rationale:** Measurement-model edge: the observed group is treated as a manifestation or measurement of the parent proxy state. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M32: `LAT_METABOLIC_DERANGEMENT` → `OUT_MORTALITY` *Dataset:* MIMIC-III **Rationale:** Outcome edge: the parent state is assumed to represent pathophysiologic burden related to mortality. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M33: `LAT_NEUROLOGIC_DYSFUNCTION` → `OBS_GCS_RASS` *Dataset:* MIMIC-III **Rationale:** Measurement-model edge: the observed group is treated as a manifestation or measurement of the parent proxy state. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** VERIFIED — PARTIALLY SUPPORTS THE STATED CLAIM. **Qualification:** SOFA supports GCS as a neurological measure; it does not establish all grouped observations.
M34: `LAT_NEUROLOGIC_DYSFUNCTION` → `OUT_MORTALITY` *Dataset:* MIMIC-III **Rationale:** Outcome edge: the parent state is assumed to represent pathophysiologic burden related to mortality. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** SOFA describes organ dysfunction and mortality association, not directed causality for this edge.
M35: `LAT_RENAL_DYSFUNCTION` → `LAT_METABOLIC_DERANGEMENT` *Dataset:* MIMIC-III **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** `O_KDIGO_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/staging-criteria-for-acute-kidney-injury-kdigo-2012. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.

Table 2 continued

Edge, rationale, provenance, independent check, and qualification
M36: `LAT_RENAL_DYSFUNCTION` → `OBS_CREATININE_BUN_URINE` *Dataset:* MIMIC-III **Rationale:** Measurement-model edge: the observed group is treated as a manifestation or measurement of the parent proxy state. **Original ChatGPT source IDs:** `O_KDIGO_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/staging-criteria-for-acute-kidney-injury-kdigo-2012. **Independent check/verdict:** VERIFIED — PARTIALLY SUPPORTS THE STATED CLAIM. **Qualification:** Sources support creatinine/urine measures, not every grouped observation or direction.
M37: `LAT_RENAL_DYSFUNCTION` → `OUT_MORTALITY` *Dataset:* MIMIC-III **Rationale:** Outcome edge: the parent state is assumed to represent pathophysiologic burden related to mortality. **Original ChatGPT source IDs:** `O_KDIGO_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/staging-criteria-for-acute-kidney-injury-kdigo-2012. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** SOFA describes organ dysfunction and mortality association, not directed causality for this edge.
M38: `LAT_RENAL_DYSFUNCTION` → `TRT_DIALYSIS` *Dataset:* MIMIC-III **Rationale:** Care-process edge: evidence of the parent state is assumed to prompt or indicate the recorded treatment. **Original ChatGPT source IDs:** `O_KDIGO_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/staging-criteria-for-acute-kidney-injury-kdigo-2012. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M39: `LAT_RESPIRATORY_FAILURE` → `LAT_METABOLIC_DERANGEMENT` *Dataset:* MIMIC-III **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** `O_BERLIN_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/berlin-definition-of-ards. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M40: `LAT_RESPIRATORY_FAILURE` → `OBS_OXYGENATION` *Dataset:* MIMIC-III **Rationale:** Measurement-model edge: the observed group is treated as a manifestation or measurement of the parent proxy state. **Original ChatGPT source IDs:** `O_BERLIN_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/berlin-definition-of-ards. **Independent check/verdict:** VERIFIED — PARTIALLY SUPPORTS THE STATED CLAIM. **Qualification:** Sources support oxygenation measurements, but Berlin requires additional diagnostic conditions.

Table 2 continued

Edge, rationale, provenance, independent check, and qualification
M41: `LAT_RESPIRATORY_FAILURE` → `OUT_MORTALITY` *Dataset:* MIMIC-III **Rationale:** Outcome edge: the parent state is assumed to represent pathophysiologic burden related to mortality. **Original ChatGPT source IDs:** `O_BERLIN_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/berlin-definition-of-ards. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** SOFA describes organ dysfunction and mortality association, not directed causality for this edge.
M42: `LAT_RESPIRATORY_FAILURE` → `TRT_MECH_VENT` *Dataset:* MIMIC-III **Rationale:** Care-process edge: evidence of the parent state is assumed to prompt or indicate the recorded treatment. **Original ChatGPT source IDs:** `O_BERLIN_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/berlin-definition-of-ards. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M43: `LAT_SHOCK` → `LAT_HEPATIC_COAG_DYSFUNCTION` *Dataset:* MIMIC-III **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M44: `LAT_SHOCK` → `LAT_METABOLIC_DERANGEMENT` *Dataset:* MIMIC-III **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M45: `LAT_SHOCK` → `LAT_RENAL_DYSFUNCTION` *Dataset:* MIMIC-III **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.

Table 2 continued

Edge, rationale, provenance, independent check, and qualification
M46: `LAT_SHOCK` → `OBS_BLOOD_PRESSURE` *Dataset:* MIMIC-III **Rationale:** Measurement-model edge: the observed group is treated as a manifestation or measurement of the parent proxy state. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** VERIFIED — PARTIALLY SUPPORTS THE STATED CLAIM. **Qualification:** Sources support blood-pressure/lactate or cardiovascular criteria, not a complete causal measurement model.
M47: `LAT_SHOCK` → `OBS_LACTATE` *Dataset:* MIMIC-III **Rationale:** Measurement-model edge: the observed group is treated as a manifestation or measurement of the parent proxy state. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** VERIFIED — PARTIALLY SUPPORTS THE STATED CLAIM. **Qualification:** Sources support blood-pressure/lactate or cardiovascular criteria, not a complete causal measurement model.
M48: `LAT_SHOCK` → `OUT_MORTALITY` *Dataset:* MIMIC-III **Rationale:** Outcome edge: the parent state is assumed to represent pathophysiologic burden related to mortality. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** The source reports mortality risk/association; observational evidence does not establish the project arrow.
M49: `LAT_SHOCK` → `TRT_VASOPRESSORS` *Dataset:* MIMIC-III **Rationale:** Care-process edge: evidence of the parent state is assumed to prompt or indicate the recorded treatment. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M50: `MISS_MEASUREMENT_INTENSITY` → `OBS_AVAILABILITY` *Dataset:* MIMIC-III **Rationale:** Observation-process edge: ordering or measurement intensity determines recorded availability/counts. **Original ChatGPT source IDs:** `O_SISK2021`. **DOI/stable URL:** doi:10.1093/jamia/ocaa242. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** Paper shows care-generated observation patterns can be predictive; it does not establish this dataset-specific arrow.

Table 2 continued

Edge, rationale, provenance, independent check, and qualification
M51: `MISS_MEASUREMENT_INTENSITY` → `OBS_LAB_COUNTS` *Dataset:* MIMIC-III **Rationale:** Observation-process edge: ordering or measurement intensity determines recorded availability/counts. **Original ChatGPT source IDs:** `O_SISK2021`. **DOI/stable URL:** doi:10.1093/jamia/ocaa242. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** Paper shows care-generated observation patterns can be predictive; it does not establish this dataset-specific arrow.
M52: `MISS_MEASUREMENT_INTENSITY` → `OBS_VITAL_COUNTS` *Dataset:* MIMIC-III **Rationale:** Observation-process edge: ordering or measurement intensity determines recorded availability/counts. **Original ChatGPT source IDs:** `O_SISK2021`. **DOI/stable URL:** doi:10.1093/jamia/ocaa242. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** Paper shows care-generated observation patterns can be predictive; it does not establish this dataset-specific arrow.
M53: `TRT_ANTIBIOTICS` → `OBS_CULTURES_MEDICATIONS` *Dataset:* MIMIC-III **Rationale:** Treatment/recording edge: delivered care changes or generates the recorded observed-data group. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M54: `TRT_DIALYSIS` → `OBS_CREATININE_BUN_ELECTROLYTES` *Dataset:* MIMIC-III **Rationale:** Treatment/recording edge: delivered care changes or generates the recorded observed-data group. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M55: `TRT_MECH_VENT` → `OBS_OXYGENATION` *Dataset:* MIMIC-III **Rationale:** Treatment/recording edge: delivered care changes or generates the recorded observed-data group. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.

Table 2 continued

Edge, rationale, provenance, independent check, and qualification
M56: `TRT_MECH_VENT` → `OBS_VENTILATOR_SETTINGS` *Dataset:* MIMIC-III **Rationale:** Treatment/recording edge: delivered care changes or generates the recorded observed-data group. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
M57: `TRT_VASOPRESSORS` → `OBS_BLOOD_PRESSURE` *Dataset:* MIMIC-III **Rationale:** Treatment/recording edge: delivered care changes or generates the recorded observed-data group. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.

4.2 PhysioNet 2012: 45 directed edges

Table 3: PhysioNet 2012 DAG-edge evidence (45 edges).

Edge, rationale, provenance, independent check, and qualification
P01: `BG_Age` → `LAT_CHRONIC_BASELINE_RISK` *Dataset:* PhysioNet 2012 **Rationale:** Project assumption: baseline/case-mix context precedes and influences the downstream proxy state. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
P02: `BG_Gender` → `LAT_CHRONIC_BASELINE_RISK` *Dataset:* PhysioNet 2012 **Rationale:** Project assumption: baseline/case-mix context precedes and influences the downstream proxy state. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
P03: `BG_HeightWeightBMI` → `LAT_CHRONIC_BASELINE_RISK` *Dataset:* PhysioNet 2012 **Rationale:** Project assumption: baseline/case-mix context precedes and influences the downstream proxy state. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
P04: `BG_ICUType` → `LAT_CARDIAC_INJURY_STRAIN` *Dataset:* PhysioNet 2012 **Rationale:** Project assumption: baseline/case-mix context precedes and influences the downstream proxy state. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
P05: `BG_ICUType` → `LAT_CHRONIC_BASELINE_RISK` *Dataset:* PhysioNet 2012 **Rationale:** Project assumption: baseline/case-mix context precedes and influences the downstream proxy state. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.

Table 3 continued

Edge, rationale, provenance, independent check, and qualification
P06: `BG_ICUType` → `MISS_MeasurementIntensity` *Dataset:* PhysioNet 2012 **Rationale:** Project assumption: case mix and care setting influence monitoring or test-ordering intensity. **Original ChatGPT source IDs:** `O_JMIR2025`. **DOI/stable URL:** https://medinform.jmir.org/2025/1/e79307. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
P07: `LAT_CARDIAC_INJURY_STRAIN` → `LAT_SHOCK` *Dataset:* PhysioNet 2012 **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
P08: `LAT_CARDIAC_INJURY_STRAIN` → `MISS_TroponinOrdering` *Dataset:* PhysioNet 2012 **Rationale:** Observation-process edge: the parent state is assumed to influence selective ordering or monitoring. **Original ChatGPT source IDs:** `O_JMIR2025`. **DOI/stable URL:** https://medinform.jmir.org/2025/1/e79307. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
P09: `LAT_CARDIAC_INJURY_STRAIN` → `OBS_TroponinHR` *Dataset:* PhysioNet 2012 **Rationale:** Measurement-model edge: the observed group is treated as a manifestation or measurement of the parent proxy state. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
P10: `LAT_CARDIAC_INJURY_STRAIN` → `OUT_InHospitalMortality` *Dataset:* PhysioNet 2012 **Rationale:** Outcome edge: the parent state is assumed to represent pathophysiologic burden related to mortality. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.

Table 3 continued

Edge, rationale, provenance, independent check, and qualification
P11: `LAT_CHRONIC_BASELINE_RISK` → `LAT_GLOBAL_SEVERITY` *Dataset:* PhysioNet 2012 **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
P12: `LAT_CHRONIC_BASELINE_RISK` → `OUT_InHospitalMortality` *Dataset:* PhysioNet 2012 **Rationale:** Outcome edge: the parent state is assumed to represent pathophysiologic burden related to mortality. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
P13: `LAT_COAG_HEME_DYSFUNCTION` → `OBS_CBCPlatelets` *Dataset:* PhysioNet 2012 **Rationale:** Measurement-model edge: the observed group is treated as a manifestation or measurement of the parent proxy state. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** VERIFIED — PARTIALLY SUPPORTS THE STATED CLAIM. **Qualification:** Sources support selected laboratory domains, not the exact project grouping.
P14: `LAT_COAG_HEME_DYSFUNCTION` → `OUT_InHospitalMortality` *Dataset:* PhysioNet 2012 **Rationale:** Outcome edge: the parent state is assumed to represent pathophysiologic burden related to mortality. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** SOFA describes organ dysfunction and mortality association, not directed causality for this edge.
P15: `LAT_GLOBAL_SEVERITY` → `MISS_MeasurementIntensity` *Dataset:* PhysioNet 2012 **Rationale:** Observation-process edge: the parent state is assumed to influence selective ordering or monitoring. **Original ChatGPT source IDs:** `O_JMIR2025`. **DOI/stable URL:** https://medinform.jmir.org/2025/1/e79307. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.

Table 3 continued

Edge, rationale, provenance, independent check, and qualification
P16: `LAT_GLOBAL_SEVERITY` → `OUT_InHospitalMortality` *Dataset:* PhysioNet 2012 **Rationale:** Outcome edge: the parent state is assumed to represent pathophysiologic burden related to mortality. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** SOFA describes organ dysfunction and mortality association, not directed causality for this edge.
P17: `LAT_GLOBAL_SEVERITY` → `TRT_MechanicalVentilation` *Dataset:* PhysioNet 2012 **Rationale:** Care-process edge: evidence of the parent state is assumed to prompt or indicate the recorded treatment. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
P18: `LAT_HEPATIC_DYSFUNCTION` → `LAT_COAG_HEME_DYSFUNCTION` *Dataset:* PhysioNet 2012 **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
P19: `LAT_HEPATIC_DYSFUNCTION` → `OBS_LiverLabs` *Dataset:* PhysioNet 2012 **Rationale:** Measurement-model edge: the observed group is treated as a manifestation or measurement of the parent proxy state. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** VERIFIED — PARTIALLY SUPPORTS THE STATED CLAIM. **Qualification:** Sources support selected laboratory domains, not the exact project grouping.
P20: `LAT_INFLAMMATION_SEPSIS_BURDEN` → `LAT_COAG_HEME_DYSFUNCTION` *Dataset:* PhysioNet 2012 **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** `O_SEPSIS3`. **DOI/stable URL:** doi:10.1001/jama.2016.0287. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** Sepsis-3 supports infection-related organ dysfunction/shock concepts, not the exact directed project edge.

Table 3 continued

Edge, rationale, provenance, independent check, and qualification
P21: `LAT_INFLAMMATION_SEPSIS_BURDEN` → `LAT_GLOBAL_SEVERITY` *Dataset:* PhysioNet 2012 **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** `O_SEPSIS3`. **DOI/stable URL:** `doi:10.1001/jama.2016.0287`. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** Sepsis-3 supports infection-related organ dysfunction/shock concepts, not the exact directed project edge.
P22: `LAT_INFLAMMATION_SEPSIS_BURDEN` → `LAT_SHOCK` *Dataset:* PhysioNet 2012 **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** `O_SEPSIS3`. **DOI/stable URL:** `doi:10.1001/jama.2016.0287`. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** Sepsis-3 supports infection-related organ dysfunction/shock concepts, not the exact directed project edge.
P23: `LAT_INFLAMMATION_SEPSIS_BURDEN` → `OBS_TempWBCInflam` *Dataset:* PhysioNet 2012 **Rationale:** Measurement-model edge: the observed group is treated as a manifestation or measurement of the parent proxy state. **Original ChatGPT source IDs:** `O_SEPSIS3`. **DOI/stable URL:** `doi:10.1001/jama.2016.0287`. **Independent check/verdict:** VERIFIED — PARTIALLY SUPPORTS THE STATED CLAIM. **Qualification:** Sepsis-3 supports infection-related organ dysfunction/shock concepts, not the exact directed project edge.
P24: `LAT_METABOLIC_DERANGEMENT` → `OBS_MetabolicLabsABG` *Dataset:* PhysioNet 2012 **Rationale:** Measurement-model edge: the observed group is treated as a manifestation or measurement of the parent proxy state. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
P25: `LAT_METABOLIC_DERANGEMENT` → `OUT_InHospitalMortality` *Dataset:* PhysioNet 2012 **Rationale:** Outcome edge: the parent state is assumed to represent pathophysiologic burden related to mortality. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.

Table 3 continued

Edge, rationale, provenance, independent check, and qualification
P26: `LAT_NEUROLOGIC_DYSFUNCTION` → `OBS_GCS` *Dataset:* PhysioNet 2012 **Rationale:** Measurement-model edge: the observed group is treated as a manifestation or measurement of the parent proxy state. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** VERIFIED — PARTIALLY SUPPORTS THE STATED CLAIM. **Qualification:** SOFA supports GCS as a neurological measure; it does not establish all grouped observations.
P27: `LAT_NEUROLOGIC_DYSFUNCTION` → `OUT_InHospitalMortality` *Dataset:* PhysioNet 2012 **Rationale:** Outcome edge: the parent state is assumed to represent pathophysiologic burden related to mortality. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** SOFA describes organ dysfunction and mortality association, not directed causality for this edge.
P28: `LAT_RENAL_DYSFUNCTION` → `LAT_METABOLIC_DERANGEMENT` *Dataset:* PhysioNet 2012 **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
P29: `LAT_RENAL_DYSFUNCTION` → `OBS_RenalLabsUrine` *Dataset:* PhysioNet 2012 **Rationale:** Measurement-model edge: the observed group is treated as a manifestation or measurement of the parent proxy state. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** VERIFIED — PARTIALLY SUPPORTS THE STATED CLAIM. **Qualification:** Sources support creatinine/urine measures, not every grouped observation or direction.
P30: `LAT_RENAL_DYSFUNCTION` → `OUT_InHospitalMortality` *Dataset:* PhysioNet 2012 **Rationale:** Outcome edge: the parent state is assumed to represent pathophysiologic burden related to mortality. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** SOFA describes organ dysfunction and mortality association, not directed causality for this edge.

Table 3 continued

Edge, rationale, provenance, independent check, and qualification
P31: `LAT_RESPIRATORY_FAILURE` → `LAT_METABOLIC_DERANGEMENT` *Dataset:* PhysioNet 2012 **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
P32: `LAT_RESPIRATORY_FAILURE` → `OBS_RespiratoryGasExchange` *Dataset:* PhysioNet 2012 **Rationale:** Measurement-model edge: the observed group is treated as a manifestation or measurement of the parent proxy state. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** VERIFIED — PARTIALLY SUPPORTS THE STATED CLAIM. **Qualification:** Sources support oxygenation measurements, but Berlin requires additional diagnostic conditions.
P33: `LAT_RESPIRATORY_FAILURE` → `OUT_InHospitalMortality` *Dataset:* PhysioNet 2012 **Rationale:** Outcome edge: the parent state is assumed to represent pathophysiologic burden related to mortality. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** SOFA describes organ dysfunction and mortality association, not directed causality for this edge.
P34: `LAT_RESPIRATORY_FAILURE` → `TRT_MechanicalVentilation` *Dataset:* PhysioNet 2012 **Rationale:** Care-process edge: evidence of the parent state is assumed to prompt or indicate the recorded treatment. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
P35: `LAT_SHOCK` → `LAT_HEPATIC_DYSFUNCTION` *Dataset:* PhysioNet 2012 **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.

Table 3 continued

Edge, rationale, provenance, independent check, and qualification
P36: `LAT_SHOCK` → `LAT_METABOLIC_DERANGEMENT` *Dataset:* PhysioNet 2012 **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
P37: `LAT_SHOCK` → `LAT_RENAL_DYSFUNCTION` *Dataset:* PhysioNet 2012 **Rationale:** Project structural hypothesis: the parent physiological state is upstream of the child state. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
P38: `LAT_SHOCK` → `MISS_LactateABGOrdering` *Dataset:* PhysioNet 2012 **Rationale:** Observation-process edge: the parent state is assumed to influence selective ordering or monitoring. **Original ChatGPT source IDs:** `O_JMIR2025`. **DOI/stable URL:** https://medinform.jmir.org/2025/1/e79307. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.
P39: `LAT_SHOCK` → `OBS_Hemodynamics` *Dataset:* PhysioNet 2012 **Rationale:** Measurement-model edge: the observed group is treated as a manifestation or measurement of the parent proxy state. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** VERIFIED — PARTIALLY SUPPORTS THE STATED CLAIM. **Qualification:** Sources support blood-pressure/lactate or cardiovascular criteria, not a complete causal measurement model.
P40: `LAT_SHOCK` → `OUT_InHospitalMortality` *Dataset:* PhysioNet 2012 **Rationale:** Outcome edge: the parent state is assumed to represent pathophysiologic burden related to mortality. **Original ChatGPT source IDs:** `O_SOFA_MSD`. **DOI/stable URL:** https://www.msdmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sofa-score. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** The source reports mortality risk/association; observational evidence does not establish the project arrow.

Table 3 continued

Edge, rationale, provenance, independent check, and qualification
P41: `MISS_LactateABGOrdering` → `OBS_MetabolicLabsABG` *Dataset:* PhysioNet 2012 **Rationale:** Observation-process edge: ordering or measurement intensity determines recorded availability/counts. **Original ChatGPT source IDs:** `O_JMIR2025`. **DOI/stable URL:** https://medinform.jmir.org/2025/1/e79307. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** Paper shows care-generated observation patterns can be predictive; it does not establish this dataset-specific arrow.
P42: `MISS_LactateABGOrdering` → `OBS_RespiratoryGasExchange` *Dataset:* PhysioNet 2012 **Rationale:** Observation-process edge: ordering or measurement intensity determines recorded availability/counts. **Original ChatGPT source IDs:** `O_JMIR2025`. **DOI/stable URL:** https://medinform.jmir.org/2025/1/e79307. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** Paper shows care-generated observation patterns can be predictive; it does not establish this dataset-specific arrow.
P43: `MISS_MeasurementIntensity` → `OBS_AvailabilityCounts` *Dataset:* PhysioNet 2012 **Rationale:** Observation-process edge: ordering or measurement intensity determines recorded availability/counts. **Original ChatGPT source IDs:** `O_JMIR2025`. **DOI/stable URL:** https://medinform.jmir.org/2025/1/e79307. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** Paper shows care-generated observation patterns can be predictive; it does not establish this dataset-specific arrow.
P44: `MISS_TroponinOrdering` → `OBS_TroponinHR` *Dataset:* PhysioNet 2012 **Rationale:** Observation-process edge: ordering or measurement intensity determines recorded availability/counts. **Original ChatGPT source IDs:** `O_JMIR2025`. **DOI/stable URL:** https://medinform.jmir.org/2025/1/e79307. **Independent check/verdict:** VERIFIED — SUPPORTS ASSOCIATION, NOT DIRECTED CAUSALITY. **Qualification:** Paper shows care-generated observation patterns can be predictive; it does not establish this dataset-specific arrow.
P45: `TRT_MechanicalVentilation` → `OBS_RespiratoryGasExchange` *Dataset:* PhysioNet 2012 **Rationale:** Treatment/recording edge: delivered care changes or generates the recorded observed-data group. **Original ChatGPT source IDs:** None found. **DOI/stable URL:** None. **Independent check/verdict:** NOT VERIFIED — NO SPECIFIC SOURCE MAPPED. **Qualification:** No edge-specific locally verified source was mapped.

5 Independent full-text verification

Eight locally available sources were inspected for this build. The original SOFA paper defines six ordinal organ-dysfunction domains and records daily worst values [8]. The Berlin definition combines timing, imaging, edema-origin, PEEP/CPAP, and oxygenation requirements [1]. KDIGO defines AKI using timed creatinine changes and urine-output criteria [4]. Sepsis-3 defines sepsis as infection-related organ dysfunction and septic shock using vasopressor-dependent hypotension with lactate above 2 mmol/L [6]. These full texts support broad clinical constructs or selected clauses, not the complete local Boolean composites.

The ISTH DIC source uses platelet count, fibrin-related markers, prothrombin time, and fibrinogen in a specific algorithm [7]; it does not validate the project's mixed hepatic/coagulation or hematologic scores. The electronic-phenotyping review supports explicit rule-based EHR phenotypes and the need for validation [2]. The respiratory phenotyping paper supports time-ordered, rule-based phenotyping from structured ventilation records, not the CliniCause gas-threshold composite [3]. The missingness paper shows that care-generated observation patterns can be predictive [5]; predictive informativeness does not establish the direction of any dataset-specific DAG edge.

6 Evidence limitations

- Some final proxies have no exact paper-to-rule mapping in the preserved ChatGPT exports; cardiac and metabolic proxies are the clearest cases.
- No implemented DAG edge has a recovered edge-specific paper mapping. Family-level sources are shown only as family-level sources.
- Several papers support broad constructs or individual clauses but not the project's local cutoffs, weights, sufficient conditions, or whole-record aggregation.
- Associations between organ dysfunction, monitoring, or mortality do not establish the direction of a causal arrow.
- Several Phase 2 sources have metadata or stable URLs but no locally available full text; these sources are not marked verified.
- Current implementation inspection does not prove the exact historical input artifact or source revision that produced every archived resource.
- Sources discovered in Phase 2 are not original ChatGPT-supplied sources unless a preserved final export independently establishes that attribution.
- No complete formal clinician-adjudication and sign-off record is available.
- These rules do not make the proxies diagnoses; the arrows do not make the DAG causal ground truth; and the supplement does not make estimated effects ground-truth causal effects.

7 References

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